



ANL Staking Protocol

WHITE PAPER

An ANL staking protocol with a dual ANL + XNT reward stream

X1 Network

Version 1.0 · July 2026

1. Executive Summary

ANL Staking Protocol is a non-custodial staking protocol for ANL tokens running on **X1 Network**. A participant locks ANL for a self-declared period (minimum 7 days) and earns two reward streams: a **fixed APY paid in ANL** from a finite, dedicated pool of 200,000,000 ANL, and a **daily-accrued share of XNT rewards** generated by a dedicated X1 validator operated by the protocol team.

The protocol offers two programs: **Genesis** — with an elevated APY that depends on the time window in which the participant joined (up to 20% per year), and **Flexible** — with a flat 8% per year regardless of when you join. In both programs the staking period is declared by the participant, and the rate assigned to a position at the moment it is opened remains **immutable for the position's entire life**.

A position settles **on the day its declared period ends**: the participant claims the ANL reward and all XNT accrued to the position at the same time. Breaking a position early returns the principal in full, but all rewards are forfeited.

Protocol parameters

Parameter	Value
Network	X1 Network (x1.xyz)
Staked token	ANL
Reward tokens	ANL (fixed APY) + XNT (daily accrual)
ANL reward pool	200,000,000 ANL (finite, segregated)
Programs	Genesis (APY windows) · Flexible (8%)
Staking period	declared by the participant, minimum 7 days
Genesis APY	20% / 15% / 8% — depending on the entry window
Flexible APY	8% — regardless of the window
Daily XNT pool split	65% Genesis programs · 35% Flexible
Reward payout	ANL + XNT together, on the day the position's period ends
Minimum deposit	1 ANL
Protocol year	365 days; periods counted in full days
Production launch	pilot with a 100,000 ANL pool, after an AI-assisted audit

2. Vision and Context

X1 Network is a high-performance blockchain compatible with the SVM tooling ecosystem. Its security and throughput are provided by validators, who are paid for their work in the native XNT token — settled after every epoch, in practice daily. ANL Staking Protocol connects the community of ANL holders with the real work of a validator: instead of promising yields from unlimited emission, the protocol shares the **validator's real, daily revenue** and pays ANL rewards from a **predefined, finite pool**. It is a model that can be calculated to the end — no hidden inflation and no dependence on an inflow of new participants.

Governing principle

User capital (principal), the ANL reward pool and the XNT pool are held in three separate vaults. The program never pays principal out of the reward vault, nor rewards out of the principal vault — an invariant hard-coded into the program and covered by automated tests.

3. Reward Tokenomics

ANL rewards come from a dedicated pool of **200,000,000 ANL** deposited in the Reward Vault. The pool acts as a liquidity reservoir: every position accrues rewards at its fixed, assigned annual rate, and payouts draw the reservoir down. A finite pool means the program has a natural horizon — the protocol does not mint new ANL.

XNT rewards come from the **revenue of the X1 validator** operated by the protocol team. The validator receives XNT after every network epoch, so the protocol has them available continuously and can accrue them to positions **daily** — no waiting for annual settlements. The validator's own deposit is never paid out to participants; only generated revenue is distributed.

4. Two Reward Streams

Every position earns in two parallel streams. Both settle at the same moment — on the day the position's period ends:

Property	ANL stream	XNT stream
Source	finite 200M ANL pool	daily X1 validator revenue
Nature	fixed annual rate (APY)	proportional share of the daily pool
Accrual	continuous, by position time and rate	daily, proportional to stake
Rate guarantee	yes — immutable for the position	no — depends on validator performance
Payout	on the day the position's period ends	together with ANL, at period end



Immutable APY

The ANL rate assigned to a position at the moment of staking holds until the end of its life — regardless of how many participants join later or how the program windows change. Protocol parameter changes never apply retroactively to open positions.

5. The Genesis Program

Genesis rewards joining the protocol early. The participant declares the staking period themselves (minimum 7 days), while the APY depends on the **time window in which they joined the program** — counted from the protocol's public launch:

Entry window	Program days	APY (ANL)
Window 1 — early access	from day 0 to the end of day 30	20% per year
Window 2	from day 31 to the end of day 90	15% per year
Window 3 — standard	from day 91	8% per year

Window boundaries are unambiguous to the second: each window covers the interval from its start until the start of the next one, and switches always occur at a fixed time of 02:00 UTC. The rate is determined solely by the moment the position is opened — a position opened in Window 1 keeps 20% per year for its entire declared period, even if that period runs long after the window closes.

Example

A participant opens a Genesis position in Window 1 for 60 days with a 100,000 ANL stake. For 60 days the position accrues its ANL reward at 20% per year ($100,000 \times 20\% \times 60/365 \approx 3,288$ ANL) plus a daily share of the XNT pool. On day 60 accrual stops, and the participant claims the ANL reward and all accrued XNT in a single claim.

6. The Flexible Program

Flexible is the simpler path for participants who join at any time and want a predictable, flat rate: **8% per year regardless of the entry window**. Position mechanics are identical to Genesis: the participant declares the period (minimum 7 days), the rate is immutable, and settlement happens on the day the period ends. Flexible positions take part in the daily XNT distribution within the 35% basket.

The difference between the programs therefore comes down to the source of the rate: Genesis rewards early entry with an elevated APY, while Flexible offers a flat 8% always — including after the higher-rate Genesis windows have closed.

7. Position Lifecycle

The following rules apply equally in both programs:

- **Opening.** The participant chooses a program, an amount (min. 1 ANL) and a period (min. 7 days). The position receives the APY in force at the moment of opening — permanently.
- **Accrual.** The ANL reward grows continuously at the position's rate; the XNT reward is credited daily, proportionally to stake (Section 8).
- **Period end.** When the declared period elapses, accrual of **both** streams stops. The position earns nothing further — even if the participant delays claiming.
- **Claim.** On the day the period ends, or at any later moment, the participant claims at once: the ANL reward, the accrued XNT, and the principal. Rewards accrued by period end are never lost after it — they simply stop growing.
- **Early exit.** The participant may withdraw capital at any time before the period ends. The principal returns immediately and in full, but **all of the position's rewards are forfeited**: the ANL reward returns to the reward pool, and the accrued XNT returns to the distribution pool.
- **Multiple positions.** Every entry creates a separate position with its own rate, period and settlement.

Breaking on day 59 of 60 — plainly

Example: a 60-day position broken on day 59 returns 100% of the capital, but the ANL reward and all accrued XNT are forfeited in full. The protocol applies no partial settlement of an unfinished period — the period is either honoured or it is not. Before you break a position, check in the app how many days remain and how much reward has already accrued.

8. XNT Rewards — Daily Mechanics

The protocol's validator receives XNT after every X1 network epoch. XNT collected within a given day goes to the XNT vault and is distributed the same day among active positions under the following rules:

- **Basket split.** The daily XNT pool is divided **65% to Genesis positions** and **35% to Flexible positions**.
- **Split within a basket.** Within a basket, every active position receives a share proportional to the size of its ANL stake relative to the sum of stakes of all active positions in that basket on that day.
- **An active position** is a position whose declared period is still running. Positions past their period end (awaiting claim) do not take part in the daily distribution.
- **Accounting.** Accrued XNT is credited to the position and accumulates for its entire period. Payout happens together with the ANL reward — on the day the position's period ends.
- **Empty basket.** If a basket has no active positions on a given day, its daily share rolls over to the next day within the same basket.



The size of the daily XNT pool depends on the validator's real performance (uptime, validator stake size, network parameters) and **is not guaranteed**. The protocol distributes 100% of the validator's generated revenue — no less, no more.

Daily accrual formula

For position i in basket k (Genesis: basket share $u_k = 0.65$; Flexible: $u_k = 0.35$), the daily XNT accrual is:

$$\text{XNT}_i = P_{\text{day}} \times u_k \times \text{stake}_i / \Sigma \text{stake}_k$$

where P_{day} is the XNT pool collected by the validator that day, and Σstake_k is the sum of ANL across all active positions in the basket.

9. Architecture and Security

Three isolated vaults

Vault	Contents	Rule
Principal Vault	participants' capital	payouts only to position owners
Reward Vault (ANL)	the 200,000,000 ANL pool	payouts only as accrued rewards
XNT Vault	validator revenue	daily distribution per Section 8

- **Untouchable principal.** Capital refunds come exclusively from the Principal Vault; no one — including the administrators — can pay out a participant's capital as a “reward”.
- **Mistake protection (ClaimFirst).** The contract will not let a position close with unclaimed matured rewards — claim and payout execute in a single transaction.
- **Controlled launch.** The protocol starts paused, goes through a pilot with a 100,000 ANL pool and 24/7 monitoring before the reward pool is fully funded.
- **Audit.** The code undergoes an AI-assisted security audit, invariant testing and 24h+ fuzzing before production launch.

10. Roadmap

Period	Stage	Scope
07–08 / 2026	Implementation & testing	Rust + Anchor smart contract; 250+ test cases; invariant tests
09 / 2026	AI audit & X1 Testnet	AI-assisted security audit; full cycle on testnet; 24h+ fuzzing
10 / 2026	Mainnet pilot	100,000 ANL pilot pool; controlled opening (launch in pause); 24/7 monitoring



Period	Stage	Scope
11-12 / 2026	Full launch	funding the 200,000,000 ANL pool; staking dashboard; daily XNT accrual starts

11. Risk Factors

- **Smart-contract risk.** Despite testing, audits and a pilot, code defects cannot be ruled out. The staged rollout limits the potential scale of damage.
- **Validator risk.** The size of XNT rewards depends on the validator's continuous, correct operation and X1 network parameters; it is not guaranteed.
- **Market risk.** The value of ANL and XNT is subject to market fluctuations independent of the protocol. The fixed APY concerns token quantity, not value.
- **Network risk.** The protocol inherits the risks of the X1 Network infrastructure (outages, epoch parameter changes, upgrades).
- **ANL pool exhaustion.** The reward pool is finite; the protocol stops accepting new positions whose rewards would lack coverage in the reservoir.

12. Glossary

Term	Meaning
Position	a single stake with its own amount, rate and period
Entry window	the range of days from launch that determines the Genesis APY
Immutable APY	the rate assigned to a position at opening, unchanged to the end
XNT basket	a share of the daily XNT pool: 65% Genesis, 35% Flexible
Active position	a position whose declared period is still running
Claim	collecting ANL + XNT rewards and principal after the period ends
Early exit	withdrawing capital before the period ends; rewards are forfeited
Epoch	the X1 network settlement cycle after which the validator receives XNT

Change History



Version	Date	Changes
1.0	07 / 2026	first publication

This document is for information purposes only and does not constitute investment advice or an offer within the meaning of applicable law. APY values refer to rewards denominated in ANL; XNT rewards depend on the validator's real performance and are not guaranteed in amount. Participation in blockchain protocols involves risk, including the risk of loss of funds.